RV1103/RV1106 And RV1103B/RV1106B Instructions API Differential

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Preface

Overview

This document primarily presents the API differences between the RV1103B/RV1106B and the RV1106/RV1103 SoCs. By comparing their differences, customers can gain a more detailed understanding of the APIs of RV1103B/RV1106B and quickly develop products based on the RV1103B/RV1106B IPC SDK.

Chipset and System Support

Chip Name	Kernel Version
RV1103B/RV1106B	Linux 5.10

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Revision History
V0.0.1	CWW	2024-08-01	Initial version
V1.0.0	CWW	2024-09-20	Add considerations for the new toolchain printf

Contents

1. Explanation of Differences in Cross Compiler tool chain

The C library for the toolchain has been upgraded to version v1.0.50, with the following main modifications:

1. The size of `sizeof(time_t)` has been changed to 8 bytes, resolving the Year 2038 timestamp overflow issue.
2. The `struct input_event` no longer has a `struct timeval` member. (Use `input_event_sec/input_event_usec` for printing time.)
3. Note that when the members of `struct timeval` and `struct timespec` are 8 bytes in size, you need to use `%lld` in `printf` for printing.

```
sizeof(timeval.tv_sec)    8
sizeof(timeval.tv_usec)   8
sizeof(timespec.tv_sec)   8
sizeof(timespec.tv_nsec)  4
```

2. Explanation of Differences in Rockit Interfaces

The RV1103B/RV1106B Rockit is developed in C language (version 2.xxx.xxx), while the RV1106 uses C++ (version 1.4.xxx). The C version of Rockit is compatible with the C++ version in terms of interfaces and debugging methods. All RV1106 interfaces are applicable to RV1103B/RV1106B. However, the Rockit C version has additional parameters and APIs in its header files compared to Rockit C++, not all of which are suitable for RV1103B/RV1106B. The following additions are applicable to RV1103B/RV1106B. For detailed usage, please refer to "Rockchip_Developer_Guide_MPI.pdf".

2.1 VI New Interfaces

2.1.1 RK_MPI_VI_SetModParam

When RK_MPI_VI_SetDevBindPipe is called to set the online mode, this API can be used to set the working mode between DEV and PIPE.

2.1.2 RK_MPI_VI_GetModParam

Retrieves the working mode between DEV and PIPE.

2.1.3 IVS New Features

OD supports interfaces for obtaining results and threshold settings by region.

```
typedef struct rkIVS_OD_INFO_S {
```

```
    RK_U32 frameId;           /* Frame number */
    RK_U32 u32Flag;
    RK_U32 u32PixSum;
    // New additions
    RK_CHAR *pData;
    RK_U32 u32Size;
    RK_S32 s32Fd;
} IVS_OD_INFO_S;

typedef struct rkIVS_OD_ATTR_S {
    RK_S32  s32ODPercent; // [5, 8]
    RK_BOOL bODUserRectEnable;
    RECT_S  stODUserRect;
    // New additions
    RK_S32  s32ThreshComplexCnt; // [0, 4]
    RK_S32  s32ThreshSad; // [0, 16383]
} IVS_OD_ATTR_S;
```

2.2 MB Differences

The maximum number of MBs has been reduced from 10240 to 1024. Projects that previously requested too many MBs should be aware of this change.

2.3 TDE New Features

TDE adds A blending.

```
typedef struct rkTDE_SURFACE_S {
    MB_BLK pMbBlk; /* <Header address of a bitmap or the Y component */
    PIXEL_FORMAT_E enColorFmt; /* <Color format */
    RK_U32 u32Height; /* <Bitmap height */
    RK_U32 u32Width; /* <Bitmap width */
    COMPRESS_MODE_E enComprocessMode; /* compress type */
    // New additions
    RK_BOOL bAlphaExt1555;  /* <Whether to enable the alpha extension of an
ARGB1555 bitmap. */
    RK_U8 u8Alpha0;  /* <Values of alpha0 and alpha1, used as the ARGB1555
format */
    RK_U8 u8Alpha1;  /* <Values of alpha0 and alpha1, used as the ARGB1555
format */
} TDE_SURFACE_S;

/* Definition of blit operation options */
typedef struct rkTDE_OPT_S {
    TDE_COLORKEY_MODE_E enColorKeyMode;  /* <Colorkey mode */
    RK_U32              unColorKeyValue; /* <Colorkey value */
    MIRROR_E            enMirror;        /* <Mirror type */
    TDE_RECT_S          stClipRect;      /* <Definition of the clipping area */
    RK_U32              u32GlobalAlpha;   /* <Global alpha value */
    // New additions
    TDE_BLENDCMD_E      eBlendCmd;       /* Blend cmd */
} TDE_OPT_S;
```

2.4 RGN New Feature

RGN adds 1BPP OSD.

2.5 VPSS Updates

VPSS introduces a new GROUP queue length setting parameter for load regulation to prevent frame loss. It requires increasing the buffer size of the preceding stage.

```
typedef struct rkVPSS_GRP_ATTR_S {
    RK_U32 u32MaxW;            /* RW; Range: [64, 16384]; Width of source image.
*/
    RK_U32 u32MaxH;            /* RW; Range: [64, 16384]; Height of source
image. */
    PIXEL_FORMAT_E enPixelFormat;      /* RW; Pixel format of source image. */
    DYNAMIC_RANGE_E enDynamicRange;    /* RW; DynamicRange of source image. */
    FRAME_RATE_CTRL_S stFrameRate;       /* Grp frame rate control. */
    COMPRESS_MODE_E enCompressMode;    /* RW; Reference frame compress mode */
    // Newly added
    RK_U32 u32MaxQueue;        /* RW; Grp Max input queue length */
} VPSS_GRP_ATTR_S;
```

2.5.1 AI (Audio Input) Updates

AI (Audio Input) introduces a new channel MAP submodule and adds all parameters for the VQE/SED algorithm modules.